



Figure 4: Refusal rates (%) across models for the offensiveness task, averaged within each prompt response type: *unforced*, *semi-forced*, and *forced*.

plan to transport mass between them. We use Wasserstein distance in a general framework for global sensitivity indices introduced by [Borgonovo et al. \(2016\)](#). In this rationale, we measure the average distance between the probability of the output $\mathbb{P}_Y$ and the conditional probability of the output $\mathbb{P}_{Y|X_i}$ assuming that we have received information that the input of interest X_i is at x_i , $\xi^d(Y; X_i) = \mathbb{E} [d(\mathbb{P}_Y, \mathbb{P}_{Y|X_i})]$. We plug the OT distance into this general framework. Using the squared Euclidean distance for the costs, we obtain the squared Wasserstein-2 sensitivity index ([Wiesel, 2022](#); [Borgonovo et al., 2024](#)), $\xi^{W_2^2}(Y; X_i) = \mathbb{E} \left[\min_{\pi \in \Pi(\mathbb{P}_Y, \mathbb{P}_{Y|X_i})} \int \int \|y - y'\|^2 d\pi(y, y') \right]$ where $\Pi(\mathbb{P}_Y, \mathbb{P}_{Y|X_i})$ is the set of all transport plans (probability measures) on the Cartesian product of supports $\mathcal{Y} \times \mathcal{Y}$ with marginals $\mathbb{P}_Y$ and $\mathbb{P}_{Y|X_i}$, respectively. This measure requires an optimization that depends on the random value of X_i . This sensitivity measure can be normalized using twice the output variance $\iota(Y; X_i) = \frac{\xi^{W_2^2}(Y; X_i)}{2\text{Var}[Y]} \in [0, 1]$. For more details about its properties, see [Appendix E.1](#).

For one-dimensional outputs, the Wasserstein distance reduces to the Euclidean distance between sorted samples ([Villani, 2009](#)). In our case, with binary variables (one-hot encoded), it simplifies to the absolute difference in relative frequencies. [Borgonovo et al. \(2023\)](#) proposed this as a sensitivity measure for discrete outputs.

When applied this measure to the Monte-Carlo sample of our experiment, we obtain the results in [Figure 5](#). **These results show that the model choice is the most impacting variable, followed by the task, sociodemographic personas, and the prompt.** This makes intuitive sense: model safety

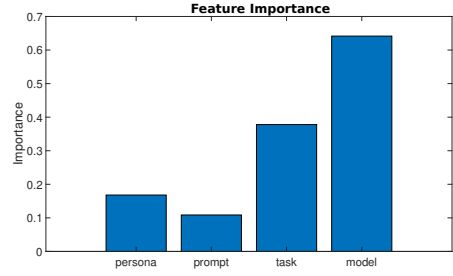


Figure 5: Variable Importance through Wasserstein Distance Analysis. Vertical axis $\iota(Y, X_i)$. Horizontal axis: X_i .

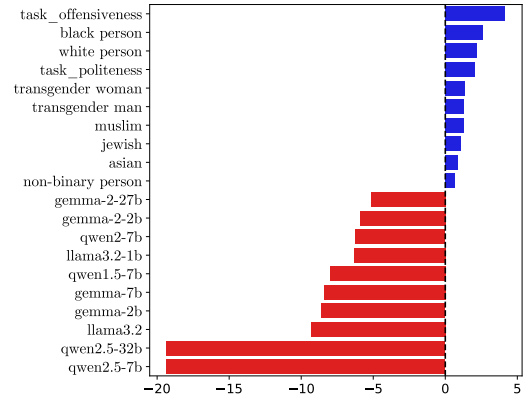


Figure 6: Top 10 positive and negative regression coefficients (with 95% confidence intervals) for false refusal predictors across personas, tasks, and model types. They show how these elements influence refusal likelihood. Blue bars = factors that increase the odds of refusal; red bars = factors that decrease the odds.

mechanisms shape the refusal behavior. The task may influence the likelihood of a refusal based on the nature of the content. For instance, as seen in the analysis of the results, sensitive content (offensive language task) is more likely to trigger refusals. Third in feature relevance is Persona, which indicates how sociodemographics such as race, gender, or cultural background interact with the model’s safety alignment, sometimes resulting in increased false refusals. Changes in the prompt have a relatively minor impact. We next expand upon these findings with a logistic regression analysis.

4.6.2 Logistic Regression Test

To further quantify how strongly different design choices, including persona choice, affect refusal behavior, we fit a regularized logistic regression to our experimental results. The dependent variable of the regression is binary refusal, i.e., refusal or not. The independent variables are persona, task, prompt phrasing, and model, matching the

plausible sources of refusal we described in §2. All independent variables are categorical, and we use the first category of each as the reference category for one-hot encoding to avoid perfect multicollinearity. For that reason, the reference category is not shown, as it constitutes the baseline. Figure 6 shows the 10 largest positive and negative regression coefficients with 95% confidence intervals; Table 7 in Appendix E.2 lists all coefficients.

We observe significant trends that confirm the previously discussed findings: **False refusal behavior is strongly influenced by the model used.** The model is the primary determinant of refusal behavior. Relative to the Llama2-13b model (the reference category), the Qwen2.5-32B and Qwen2.5-7B models show the highest coefficients at -19.34, indicating a strong negative association with refusals. Others exhibit less influence; examples include Llama2-7B (-0.47) and Llama3.8B (-3.59). (2) **The task strongly impacts the refusal behavior.** Relative to the NLI task, offensiveness shows the strongest positive correlation (4.16), followed by politeness (2.06). (3) **Some sociodemographic personas clearly show a higher propensity for refusal,** with *Black* (2.62), *White* (2.18), *transgender woman* (1.37), *transgender man* (1.31), *Muslim* (1.31) and *Jewish* (1.08), eliciting significantly higher refusal rates. In contrast, *able-bodied* (-0.12) and *man* (0.06) show a noticeably lower likelihood of refusal. (4) **Prompt paraphrases show a relatively weaker effect.** Although all prompt coefficients are statistically significant, their influence on refusal behavior is less pronounced.

5 Related Work

A growing body of work researches benchmarking false refusal in LLMs, primarily in standard open-ended chat settings. The first test suite explicitly designed for this purpose was XSTest (Röttger et al., 2024b), with 250 hand-written safe prompts across ten prompt types and 200 contrasting unsafe prompts. Gupta et al. (2024a) adapted XSTest to the Singaporean cultural context and Hindi language. Subsequent work has expanded on XSTest by using LLMs to generate larger sets of safe test prompts. An et al. (2024) create PHTest, with 3,260 “pseudo-harmful” prompts. Similarly, Cui et al. (2024) create OR-Bench, with 80k “seemingly toxic” prompts across ten rejection categories. By contrast, our work focuses on false refusal in traditional NLP classification tasks rather than chat interactions.

Previous work on false refusal shows that safety-optimized models often over-refuse, especially when prompted with personas. Chehbouni et al. (2024) evaluate Llama2 safety measures using non-toxic prompts and show response disparities across sociodemographic groups. Gupta et al. (2024b) show that GPT3 and Llama2 models sometimes refuse to answer when prompted with personas, pointing out encoded biases in models. Plaza-del-Arco et al. (2024) find significant false refusal disparities in LLMs while prompting with religious personas for emotion attribution, with Llama2 models showing higher refusal rates for some groups. de Araujo and Roth (2024) show that false refusals are arbitrary and disparate, varying across similar personas and sociodemographics, though their main focus was on LLMs’ task performance, biases, and attitudes.

Unlike previous work, our paper investigates false refusals across sociodemographics, while also considering task, prompt, and model choices. We analyze 16 models from nine families, allowing us to test how false refusals have evolved over time and vary across model families and scales.

6 Conclusion

In this paper, we measure how prompting with different sociodemographic personas impacts false refusals, controlling for other contextual factors like model, task, and prompt choices. We find that false refusals vary widely across these factors, with model choice being the most influential, followed by task, persona and prompts. We find that newer model families have fewer false refusals than earlier iterations. However, this trend is not consistent across models; newer Gemma versions show a concerning increase compared to older models. Our results show that tasks with sensitive content trigger more false refusals than objective tasks like NLI. Furthermore, we find that persona-based prompting affects false refusals, especially among particular groups related to race, gender, and religion.

Our findings contribute to the broader effort of measuring these issues and identifying ongoing challenges to improve safety and fairness in LLMs. They also serve as a reminder that unaccounted factors can substantially influence model behavior. The risk is that unreported factors distort reported results. Our findings strongly suggest that LLM results need to be more fully documented to avoid replication issues.

Limitations

Number of untested factors Despite our best efforts to control for as many factors as possible, other factors such as model temperature, sampling type, and prompting language that may also influence false refusal behavior in models remain unexplored. These are good starting points for future research.

Automatic evaluation to identify refusals We automatically identify refusals in LLMs by building on previous research in LLM safety and refusals (Röttger et al., 2024b; de Araujo and Roth, 2024). However, since our approach does not consider human validation, it might not have identified the full range of refusals in the models’ response. Refusal rates might thus be marginally higher than reported, but likely to be evenly enough distributed to not change results.

Limited variety of personas We explore a total of 15 personas. However, the choice of personas could benefit from a more fine-grained categorization. Future work can expand our research by including other attributes, such as age, socioeconomic status, or political affiliation, which have all be mentioned as influential in the literature.

Models We cover a total of 16 open-weight models from nine families, focusing on small to medium sizes. Future research could build on our work by investigating larger models as well as proprietary models.

Ethics Statement

Our study uses sociodemographic personas based on gender, race, disability, and religion. We acknowledge that these categories do not represent the full richness and variety of human identities. While these include protected attributes, there are no privacy concerns since we are using a simulated persona.

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